UNIT-4 HUMAN PHYSIOLOGY

BLOOD VASCULAR SYSTEM

Components of blood

Blood grouping in humans

Myogenic and neurogenic hearts

Structure of human heart

ENDOCRINE SYSTEM

List of various glands and their secretions

COMPONENTS OF BLOOD

You have learnt that all living cells have to be provided with nutrients, O_2 and other essential substances. Also, the waste or harmful substances produced have to be removed continuously for healthy functioning of tissues. It is therefore, essential to have efficient mechanisms for the movement of these substances to the cells and from the cells. Different groups of animals have evolved different methods for this transport. Simple organisms like sponges and coelenterates circulate water from their surroundings through their body cavities to facilitate the cells to exchange these substances. More complex organisms use special fluids within their bodies to transport such materials. Blood is the most commonly used body fluid by most of the higher organisms including humans for this purpose. Another body fluid, lymph, also helps in the transport of certain substances. In this chapter, you will learn about the composition and properties of blood and lymph (tissue fluid) and the mechanism of circulation of blood is also explained herein.

BLOOD

Blood is a special connective tissue consisting of a fluid matrix, plasma and formed elements.

PLASMA

Plasma is a straw coloured, viscous fluid constituting nearly **55 per cent** of the blood. 90-92 per cent of plasma is **water** and proteins contribute 6-8 per cent of it. Albumins, globulins and Fibrinogens are the major proteins. Albumins help in osmotic balance, globulins primarily are involved in defense mechanisms of the body and fibrinogens are needed for clotting or coagulation of blood. Plasma also contains small amounts of minerals like **Na^+, Ca^{++}, Mg^{++}, HCO_3^-, Cl^-, etc. Glucose, amino acids, lipids**, etc., arenalso present in the plasma as they are always in transit in the body. Factors for coagulation or clotting of blood are also present in the plasma in an inactive form. Plasma without the clotting factors is called **serum**.

FORMED ELEMENTS

Erythrocytes, leucocytes and platelets are collectively called formed elements and they constitute nearly 45 per cent of the blood. These include:

ERYTHROCYTES or red blood cells (RBC) are the most abundant of all the cells in blood. A healthy adult man has, on an average, **5 million to 5.5 millions** of RBCs mm^{-3} of blood. RBCs are formed in the **red bone marrow** in the adults. RBCs are devoid of nucleus in most of the mammals and are biconcave in shape. They have a red coloured, iron containing complex protein called haemoglobin, hence the colour and name of these cells. A healthy individual has 12-16 gms of haemoglobin in every100 ml of blood. These molecules play a significant role in transport of respiratory gases. RBCs have an average life span of 120 days after which they are destroyed in the spleen (graveyard of RBCs).

LEUCOCYTES are also known as white blood cells (WBC) as they are colourless due to the lack of haemoglobin. They are nucleated and are relatively lesser in number which averages 6000-8000 mm^{-3} of blood. Leucocytes are generally short lived. We have two main categories of WBCs

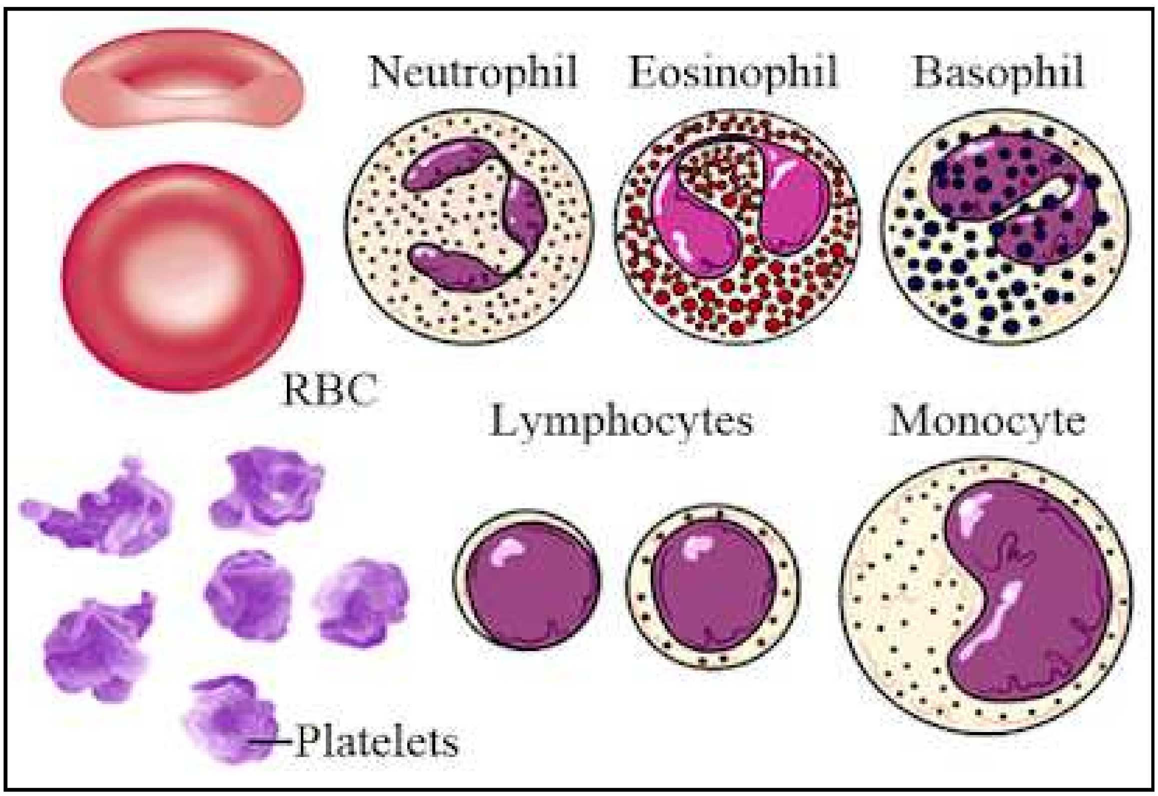
1) **Granulocytes**: basophils, eosinophils and neutrophils are different types of granulocytes. Basophils (0.5-1 per cent) secrete heparin, histamine, serotonin, etc., and are involved in inflammatory reactions. Eosinophils (2-3 per cent) resist infections and are also associated with allergic reactions. Neutrophils are the most abundant cells (60-65 per cent) of the total WBCs. They are phagocytic cells which destroy foreign organisms entering the body.

2) **Agranulocytes:** lymphocytes and monocytes are the agranulocytes.

   Lymphocytes (20-25 per cent) which include B and T lymphocytes are responsible for immune responses of the body.

Monocytes (7%) are phagocytic cells which destroy foreign organisms entering the body. They are also responsible for removing cellular debris and any other foreign substance that may find its way into the bloodstream thus acting as a scavenger.

PLATELETS also called thrombocytes, are cell fragments produced from megakaryocytes (special cells in the bone marrow). Blood normally contains 1,500,00 - 3,500,00 platelets mm^{-3}. Platelets can release a variety of substances most of which are involved in the coagulation or clotting of blood. A reduction in their number can lead to clotting disorders which will lead to excessive loss of blood from the body.



BLOOD GROUPS

As you know, bloods of human beings differ in certain aspects though it appears to be similar. Various types of grouping of blood has been done.

Two such groupings – the ABO and Rh – are widely used all over the world.

ABO grouping

ABO grouping is based on the presence or absence of two surface antigens (chemicals that can induce immune response) on the RBCs namely A and B. Similarly, the plasma of different individuals contain two natural antibodies (proteins produced in response to antigens). The distribution of antigens and antibodies in the four groups of blood, A, B, AB and O are given in below Table. You probably know that during blood transfusion, any blood cannot be used; the blood of a donor has to be carefully matched with the blood of a recipient before any blood transfusion to avoid severe problems of clumping (destruction of RBC). The donor's compatibility is also shown in the below Table. From the above mentioned table it is evident that group 'O' blood can be donated to persons with any other blood group and hence 'O' group individuals are called '**universal donors**'. Persons with 'AB' group can accept blood from persons with AB as well as the other groups of blood. Therefore, such persons are called '**universal recipients**'.

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

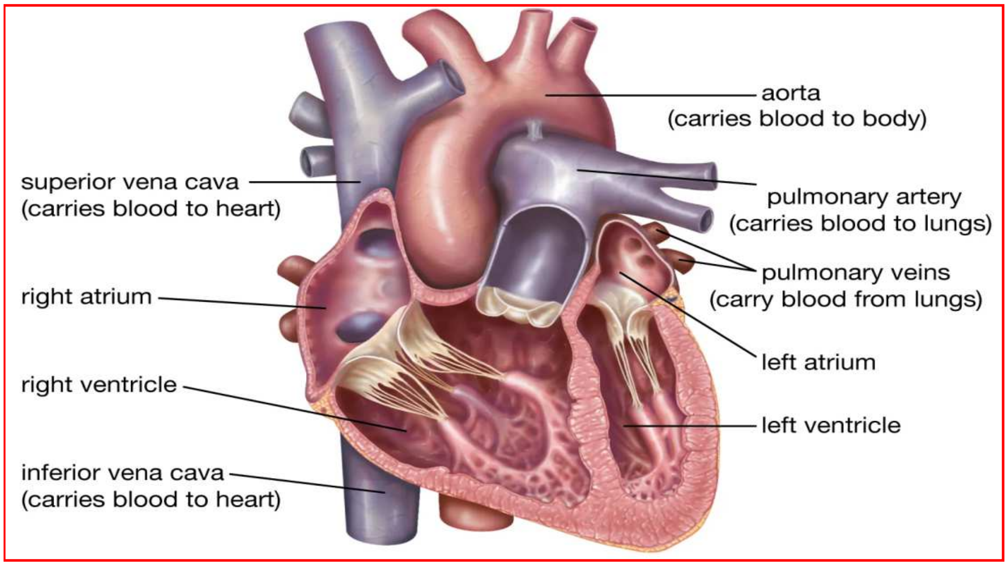
Rh grouping

Another antigen, the **RhD** antigen similar to one present in Rhesus monkeys (hence Rh), is also observed on the surface of RBCs of majority (nearly 80 per cent) of humans. Such individuals are called Rh positive (Rh+ve) and those in whom this antigen is absent are called Rh negative (Rh-ve).

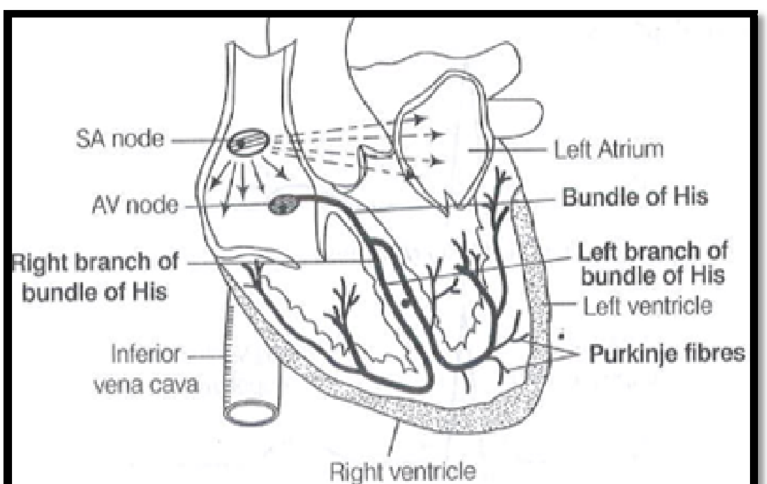
STRUCTURE OF HUMAN HEART

Heart, the **mesodermally** derived organ, is situated in the thoracic cavity, in between the two lungs, slightly tilted to the left. It has the size of a clenched fist. It is protected by a double walled membranous bag, **pericardium**, enclosing the **pericardial fluid**. Our heart has **four chambers**, two relatively small upper chambers called **atria** which are thin walled receiving chambers and two larger lower chambers called **ventricles** which are thick walled distributing chambers. A thin, muscular wall called the **interatrial septum** separates the right and the left atria, whereas a thick-walled, the **inter-ventricular septum**, separates the left and the right ventricles. The atrium and the ventricle of the same side are also separated by a thick fibrous tissue called the **atrio-ventricular septum**.

However, each of these septa is provided with an opening through which the two chambers of the same side are connected. The opening between the right atrium and the right ventricle is guarded by a valve formed of three muscular flaps or cusps, the tricuspid valve, whereas a bicuspid or mitral valve guards the opening between the left atrium and the left ventricle. The openings of the right and the left ventricles into the pulmonary artery and the left systemic aorta respectively are provided with the semilunar valves. The valves in the heart allows the flow of blood only in one direction, i.e., from the atria to the ventricles and from the ventricles to the pulmonary artery or aorta. These valves prevent any backward flow. The right side of the heart receives deoxygenated blood and the left side of the heart receives oxygenated blood. Right atrium receives deoxygenated blood through superior vena cava (precaval vein) and inferior venacava (postcaval vein). The left atrium receives oxygenated blood through pulmonary veins. These are not guarded by valves.



The entire heart is made of **cardiac muscles**. The walls of ventricles are much thicker than that of the atria. A specialized cardiac musculature called the **nodal tissue** is also distributed in the heart. A patch of this tissue is present in the **right upper corner** of the right atrium called the **sino-atrial node (SAN) or pacemaker**. Another mass of this tissue is seen in the **lower left corner** of the right atrium close to the atrio-ventricular septum called the atrio-ventricular node (AVN). A bundle of nodal fibres, Bundle of His or atrioventricular bundle (AV bundle) continues from the AVN which passes through the atrio-ventricular septa to emerge on the top of the interventricular septum and immediately divides into a right and left bundle. These branches give rise to minute fibres throughout the ventricular musculature of the respective sides and are called purkinje fibres. The nodal musculature has the ability to generate action potentials without any external stimuli, i.e., it is **autoexcitable**.



Types of heart

There are two types of hearts Neurogenic and myogenic hearts. A neurogenic Heart is a heart that beats by nerve impulses. The myogenic heart is the heart that beats by specialized muscle cells.

Neurogenic Heart	**Myogenic heart**
It is the characteristic of lower invertebrates.	It is the characteristic of vertebrates.
It is an open circulatory system.	It is a close circulatory system.
The structure of the heart is sac-like or tubular.	The structure of the heart comprises 2-4 chambers.
A group of nerve impulses initiates the heartbeat.	A Group of muscle cells initiates a heartbeat.

MULTIPLE CHOICE QUESTIONS

1) Percentage of plasma in blood is
   a) 70 – 80%
   b) **55%**
   c) 20%
   d) 10%

2) The average span of life of erythrocytes in human beings
   a) 200 days
   b) 50 days
   c) **120 days**
   d) 150 days

3) RBC are formed in
   a) Stomach
   b) Bone
   c) **Bone marrow**
   d) Pancreas

4) The type of leucocytes which attack and engulf bacteria in the blood are

a) **Neutrophils**
b) Eosinophils
c) Basophils
d) Monocytes

5) Highest number of WBC are of the type
a) **Neutrophils**
b) Monocytes
c) Lymphocytes
d) Basophils

6) If blood cells from blood are eliminated, the liquid left is
a) Serum
b) **Plasma**
c) Lymph
d) Almost water

7) The agranulocytes are
a) **Lymphocytes and monocytes**
b) Eosinophils and basophils
c) Lymphocytes and eosinophils
d) Basophils and Monocytes

8) When RBC count decreases, it is called
a) Poly cythemia
b) **Anaemia**
c) Leucocytosis
d) Leucopenia

9) WBC which act as scavengers and help in removing damaged tissues
a) Lymphocytes
b) **Monocytes**
c) Eosinophils
d) Basophils

10) The number of pericardial membranes that surround the heart are
a) One
b) Two
c) Three
d) Four

11) The left auricle receives oxygenated blood from lungs through
a) **Pulmonary vein**
b) Pulmonary artery

c) Pre caval vein
d) Post caval vein

12) The distributing chambers of the heart are
a) Auricles
b) Ventricles
c) Sinus venosis
d) All

13) The right auriculo-ventricular opening is guarded by
a) **Tricuspid valve**
b) Bicuspid valve
c) Eustachian valve
d) Semilunar valve

14) The precaval open into
a) **Right auricle**
b) Left auricle
c) Right ventricle
d) Left ventricle

15) The number of aortic arches arise from the ventricles in man
a) 1
b) 2
c) 3
d) 4

16) Purkinje fibres originate from
a) Sinu-auricular node
b) Auriculo-ventricular node
c) Bundle of His
d) Sinus venosus

17) The left systemic arch of man originates from
a) Right auricle
b) Right ventricle
c) Left ventricle
d) Left auricle

18) The left auricle of man receives
a) Oxygenated blood
b) Deoxygenated blood
c) Mixed blood
d) Both pure and impure blood

19) The pace maker of the heart of man is
   a) Sinuauricular node
   b) Auriculo ventricular node
   c) Bundle of His
   d) Purkinje fibres

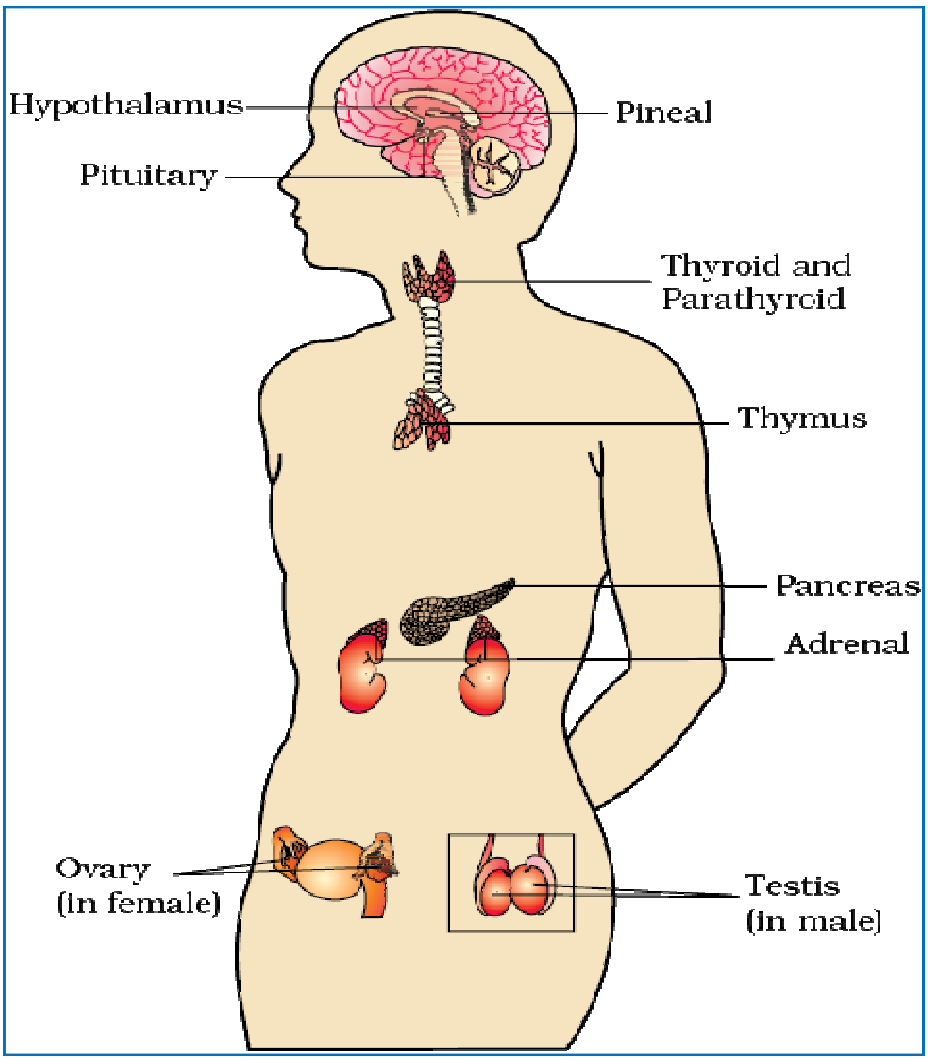
20) The valve situated between left auricle and left ventricle is

   a) Mitral valve
   b) Semilunar valve
   c) Eustachian valve
   d) None

ENDOCRINE SYSTEM

Endocrine glands and their secretions

The endocrine glands and hormone producing diffused tissues/cells located in different parts of our body constitute the endocrine system. Pituitary, pineal, thyroid, adrenal, pancreas, parathyroid, thymus and gonads (testis in males and ovary in females) are the organized endocrine bodies in our body. In addition to these, some other organs, e.g., gastrointestinal tract, liver, kidney, heart also produce hormones. A brief account of the structure and functions of all major endocrine glands and hypothalamus of the human body is given in the following sections.



THE HYPOTHALAMUS

The hypothalamus is the basal part of diencephalon, forebrain and it regulates a wide spectrum of body functions. It contains several groups of **neurosecretory cells called nuclei** which produce hormones. The hormones are:

1) **Releasing hormones** (which stimulate secretion of pituitary hormones).

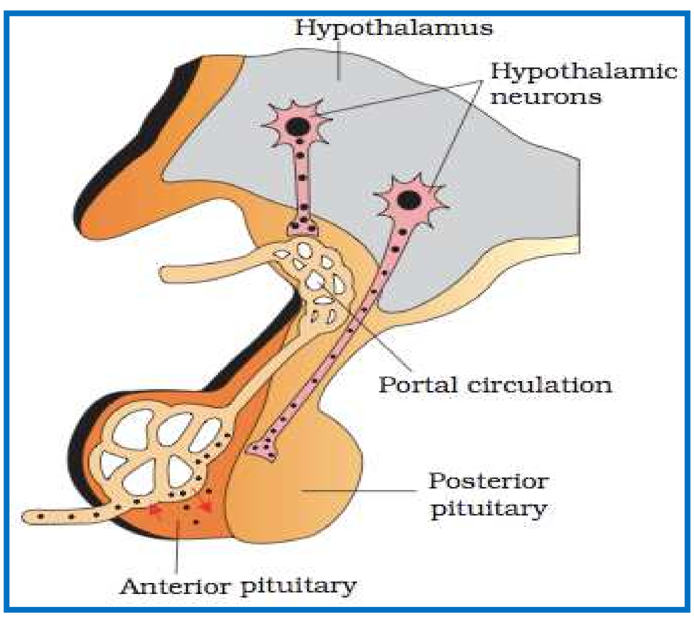
   Eg. GHRH (growth hormone-releasing hormone) which aids in growth

2) **Inhibiting hormones** (which inhibit secretions of pituitary hormones)

   Eg. GHIH that regulates muscle and bone growth.

3) **Oxytocin** (stimulate uterine contractions during childbirth and also contractions of breast tissue to aid in lactation after childbirth)

4) **Vasopressin** (Anti Diuretic Hormone, **ADH**) - has constricting effect on arterioles and also decreases water excretion by the kidneys.



THE PITUITARY GLAND

The pituitary gland is the smallest endocrine gland located in a bony cavity called sella tursica and is attached to hypothalamus by a stalk. It is called as the **master gland**. It is divided anatomically into an **adenohypophysis** and **neurohypophysis**.

1) Adenohypophysis consists of two portions

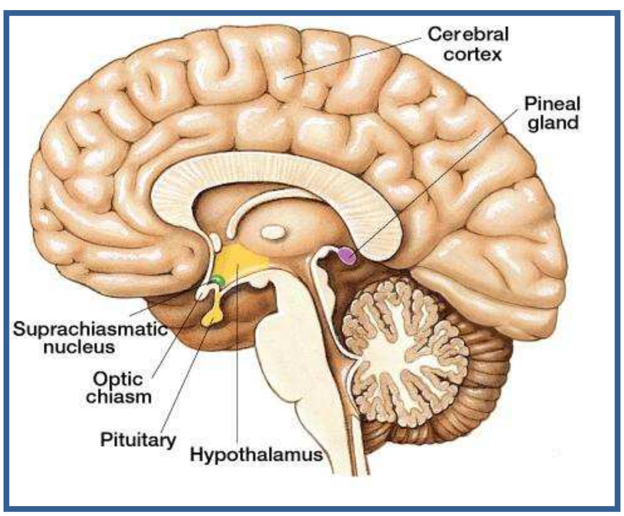
    Pars distalis (anterior pituitary) – which secretes gonadotrophic hormones, adrenocorticotrophic hormone (ACTH), thyroid stimulating hormone (TSH), prolactin (PRL), growth hormone (GH)

    Pars intermedia- secretes only one hormone called melanocyte stimulating hormone (MSH).

2) Neurohypophysis/parsnervosa (posterior pituitary) - stores and releases the hormones oxytocin and vasopressin.

THE PINEAL GLAND

The pineal gland is located on the dorsal side of forebrain. It secretes a hormone called **melatonin**. Melatonin plays a very important role in the regulation of a 24-hour (diurnal) rhythm of our body (i.e sleep-wake cycle) and body temperature.



THYROID GLAND

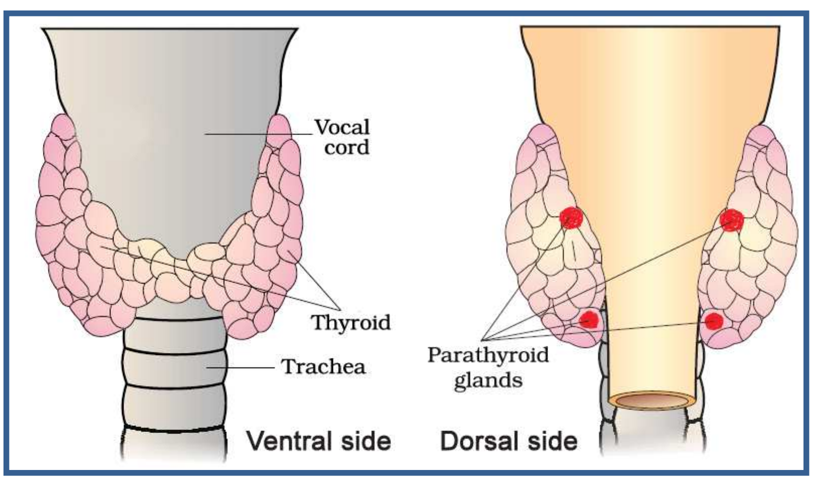
The thyroid is a butterfly-shaped gland. It is the largest endocrine gland composed of two lobes which are located on either side of the trachea. Both the lobes are interconnected with a thin flap of connective tissue called isthmus. The thyroid gland is composed of two types of endocrine cells:

1) **Follicular cells** which secrete iodinated hormones (**T3-**triiodothyronine; **T4-**tetraiodothyronine or thyroxine). Iodine is essential for the normal rate of hormone synthesis in the thyroid. These hormones control the body metabolism, growth and mental development.

2) **Parafollicular cells** (**C-cells**) secrete **Calcitonin** which decreases Ca^{2+} levels in the blood.

PARATHYROID GLAND

In humans, four parathyroid glands are present on the back side of the thyroid gland. They secrete a hormone called parathyroid hormone (PTH). When Calcium levels in blood are low PTH increases the Ca^{2+} levels in the blood by stimulating the bones to release calcium, increasing calcium absorption from digested food and by reabsorption of Ca^{2+} by the renal tubules.



THYMUS

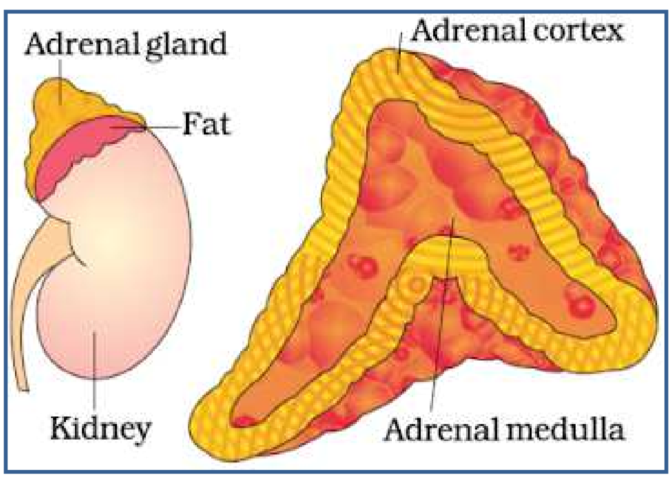
The thymus gland is a lobular structure located between lungs behind sternum on the ventral side of aorta. This gland secretes the hormones called **thymosins** which play an **important role in immunity**. Thymus is degenerated in old individuals resulting in a decreased production of thymosins. As a result, the immune responses of old persons become weak.

ADRENAL GLAND

Our body has one pair of adrenal glands, one at the anterior part of each kidney. The gland is composed of two types of tissues.

1) The **adrenal cortex** (outer tissue) secretes many hormones commonly called as **corticoids** (the **glucocorticoids** regulate carbohydrate metabolism and the **mineralocorticoids** regulate water and electrolyte balance).

2) The **adrenal medulla** (inner tissue) secretes two hormones called **adrenaline (epinephrine)** and **noradrenaline (norepinephrine)**. Both are rapidly secreted in response to stress of any kind and during emergency situations and are called **emergency hormones or hormones of Fight or Flight**.

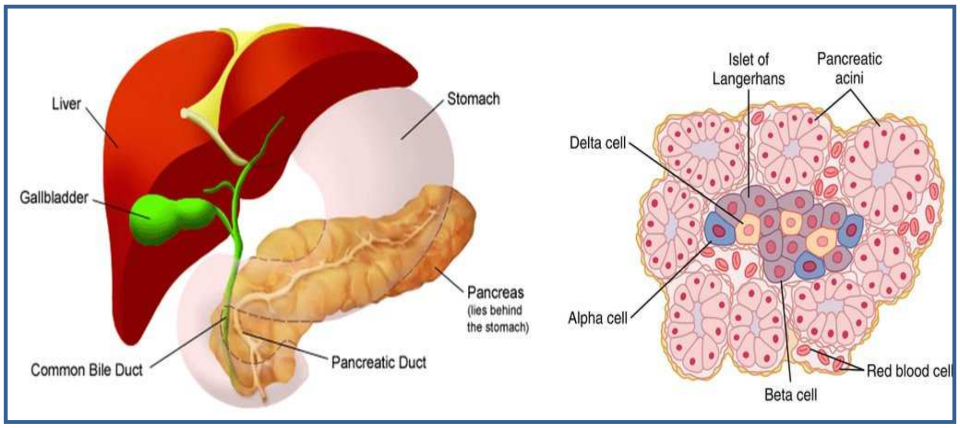


PANCREAS

Pancreas is a **composite gland** which acts as both endocrine gland and exocrine gland. The endocrine part consists of 'Islets of Langerhans' which is composed of two types of cells:

α- cells secrete glucagon which controls blood glucose levels.

β- cells secrete insulin. Insulin allows the cells of the body to use glucose as energy and to maintain the glucose within normal limit. Deficiency of insulin would lead to Diabetes mellitus.



TESTIS

A pair of testis is present in the scrotal sac (outside abdomen) of male individuals. Testis performs dual functions as a **primary sex organ as well as an endocrine gland**. Testis is composed of seminiferous tubules and interstitial cells (Leydig cells). The **interstitial cells** produce a group of hormones called **androgens mainly testosterone**. The hormone is responsible for **secondary sexual characters**.

OVARY

Females have a pair of ovaries located in the abdomen. It performs dual functions as a **primary sex organ as well as an endocrine gland**. Ovary is composed of **ovarian follicles** which secrete **estrogens** responsible for **secondary sexual characters**. After ovulation, the ruptured follicle is converted to a structure called **corpus luteum**, which secretes mainly **progesterone**. Progesterone supports **pregnancy** and **secretion of milk from mammary glands**.

MULTIPLE CHOICE QUESTIONS

21) Which of the following is not a gastro intestinal hormone
    a) **Thyroxine**
    b) Secretin
    c) Cholecystokinin
    d) Enterogestrone

22) Which cells of the islets of langerhans secrete insulin
    a) Alpha cells
    b) **Beta cells**
    c) Gama cells
    d) C – cells

23) The posterior lobe of pituitary is known as
    a) Adeno hypopihysis
    b) **Neuro hypophysis**
    c) Epiphysis
    d) Endophysis

24) The hormone which decreases water excretion by kidneys
    a) Oxytocin
    b) **Vasopressin**
    c) Prolactin
    d) Adrenaline

25) Master endocrine gland is
    a) Adrenal
    b) **Pituitary**
    c) Pancreas
    d) Pineal

26) The hormone essential for normal growth and mental development
    a) Adrenaline
    b) Oxytocin
    c) **Thyroxine**
    d) Insulin

27) The hormone which regulates the concentration of calcium in blood
    a) Thyroxine
    b) Adrenaline

c) Oxytocin
d) **Parathyroid hormone**

28) The endocrine gland associated with the kidney is
a) **Adrenal**
b) Thyroid
c) Pituitary
d) Pineal

29) The gland for fight, fright and flight is
a) Adrenal cortex
b) **Adrenal medulla**
c) Thyroid
d) Pituitary

30) Diabetes mellitus is caused by the deficiency of
a) Glycogen
b) **Insulin**
c) Oxytocin
d) ADH

31) The development of secondary sexual characters in males is by
a) **Testosterone**
b) Estrogen
c) Aldosterone
d) Epinephrine